3. Model Performance

This section reports how the classification and regression models performed on the held-out test set (50,000 properties, 20% of the data). All numbers come from `reports/metrics.json` and the notebook outputs. The models were trained on the remaining 200,000 properties.

3.1 Classification: Good Investment

Comparison of models

Model	Accuracy	F1 score	ROC-AUC
Logistic Regression	0.7081	0.7071	0.7794
Random Forest (selected)	**0.7083**	**0.7093**	0.7729
XGBoost	0.7075	0.7086	0.7774

Selection rule: the model with the highest F1 score was selected. Random Forest was selected with an F1 score of 0.7093.

All three models perform almost identically (accuracy between 70.75% and 70.83%, F1 between 0.707 and 0.709). The differences are too small to show that one model is truly better than the others. Logistic Regression has the highest ROC-AUC (0.7794), which means it ranks properties slightly better, but it was not selected because the selection rule was based on F1.

Classification report (Random Forest, test set)

Class	Precision	Recall	F1 score	Support
Not good (0)	0.71	0.70	0.71	25,003
Good (1)	0.71	0.71	0.71	24,997
Overall accuracy			**0.71**	50,000

Precision and recall are balanced between the two classes, so the model does not favour one label over the other.

Confusion matrix (Random Forest, test set)

	Predicted: Not good	Predicted: Good
Actual: Not good	17,625 (correct)	7,378 (wrong)
Actual: Good	7,207 (wrong)	17,790 (correct)

- Total correct predictions: 35,415 out of 50,000 (70.8%).
- Total wrong predictions: 14,585 (29.2%).
- Errors are split evenly between the two directions, so the model does not show a bias towards one label.

Interpretation

- A random guess on this balanced data would be right about 50% of the time. The model is right about 71% of the time, which is a real improvement, but not a strong one.
- The ceiling is limited by the labelling rule. The label depends on each property's price per sq ft relative to its city median. The model cannot see price columns, so it has to estimate this from other features, and it does so with limited accuracy.
- Accuracy around 70% is a reasonable result for this type of problem. It should not be presented as a reliable investment advisor. The model is best described as a tool that highlights which properties are likely to be cheap for their city, based on size, location and other features.

Feature importance (Random Forest classifier)

Rank	Feature	Importance
1	Size_in_SqFt	0.677
2	Floor_No	0.027
3	Total_Floors	0.027
4	Year_Built	0.023
5	Age_of_Property	0.023
6	Nearby_Hospitals	0.019
7	Nearby_Schools	0.018
8	Amenities_Count	0.014
9	BHK	0.014
Rest	Parking, Security, Owner type and others	below 0.006 each

Size_in_SqFt accounts for about 68% of the total importance. Every other feature has a small share. This is the most important finding of the classification model. Because price per sq ft falls as size rises in this dataset, the model mostly predicts "Good Investment" for larger properties and "Not good" for smaller ones. Section 4 explains why this is a limitation and not a valid investment rule.

3.2 Regression: 5-year future price

Comparison of models

Model	RMSE (lakh)	MAE (lakh)	R^2
Fixed-rate baseline (8% per year for all)	9.55	2.59	0.9979
Linear Regression	0.0002	0.0001	1.0000
Random Forest (selected)	**0.0000**	**0.0000**	**1.0000**
XGBoost	0.0078	0.0024	1.0000

Selection rule: the model with the lowest RMSE was selected. Random Forest had the lowest RMSE.

How to read these results

The R^2 values close to 1.0 and the very low errors **do not show that the models predict the real market well.** The dataset has no actual future prices. The target was calculated with a fixed formula:

> Future price = Current price × (1 + city growth rate)^5

The city growth rates are assumptions made for this project (see Section 1.4). The models learned the growth rate for each city from the data and then reproduced the formula almost exactly. This is why the errors are so small.

The more useful comparison is between the **fixed-rate baseline** and the **ML models**:

- The fixed 8% rule, which ignores city, has an RMSE of 9.55 lakh.
- The ML models, which learned the city-specific rates, have an RMSE close to zero.

This result shows that the models can recover a pattern that is built into the data. It does not show that they can forecast real house prices.

Feature importance (Random Forest regressor)

Rank	Feature	Importance
1	City: Kolkata	0.413
2	City: Bangalore	0.301
3	City: Hyderabad	0.111
4	City: Chennai	0.103
5	City: Mumbai	0.072

City features make up almost all of the importance. This is expected, because the growth rate is set per city. Price and size have almost no importance, because the model only needs to learn the growth ratio, not the price itself.

3.3 Model selection summary

Task	Selected model	Main metric	Score on test set
Good Investment (classification)	Random Forest	F1 score	0.7093
5-year price (regression)	Random Forest	RMSE	≈ 0 lakh (rule-based target)

Both selected models are Random Forests. They were trained with the same settings (100 trees, max depth 15, minimum 5 samples per leaf). The Random Forest classifier was

trained on a sample of 60,000 rows, and the regressor on the same sample, to keep training time reasonable.

3.4 Overfitting check

Overfitting means the model performs much better on the training data than on new data. The notebook includes a check that compares training and test scores.

Measure	Training score	Test score	Gap
Classifier F1 (Random Forest)	0.7203	0.7093	0.0110 (about 1.1 points)
Regressor R^2 (Random Forest)	1.0000	1.0000	0.0000

A gap of 2 to 3 F1 points or less is generally considered acceptable. A larger gap would indicate overfitting.

Result:

- **Classifier:** The training F1 (0.7203) is about 1.1 points higher than the test F1 (0.7093). This is within the acceptable range, so the Random Forest shows only mild overfitting. The model is learning patterns that generalise to new data, although it does memorise the training set slightly more than it does the test set.
- **Regressor:** The R^2 is 1.0 on both training and test sets. This is expected, because the target is a fixed formula. It does not indicate overfitting, but it also does not show real forecasting skill (see Section 3.2).

The training score was measured on all 200,000 training rows, while the Random Forest was fitted on a 60,000-row sample. Because the evaluation rows include rows the model did not see during fitting, the gap reported here is a fair measure of generalisation.

3.5 Model artifacts

- The trained models are saved as `models/classifier.joblib` and `models/regressor.joblib`.
- Both models are registered in the MLflow Model Registry (see Section 1.9).
- The full set of metrics is in `reports/metrics.json`.
- The Streamlit app loads these saved models to make predictions for new properties.